

Vision-Only Reactive Corridor Navigation for a Quadruped: Reusing a Frozen Velocity Policy with a Hard Passage Gate

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Video: github.com/teshnizi2/go2-ball-follower/blob/main/channel/demo/go2_submission.mp4

Code: github.com/teshnizi2/go2-ball-follower

Abstract. A simulated Unitree Go2 quadruped follows a moving red ball down a cluttered corridor using only its head camera: no GPS, lidar, depth sensor, or map. The gait comes from a frozen, pre-trained reinforcement-learning policy whose weights we never touch; around it we build a colour tracker, a reactive gap-picking planner, and the corridor generator. The policy accepts a sideways velocity command, but a controller that only walks toward a target leaves that channel at zero, so the robot turns to move sideways, like a car changing lanes. We switch the channel back on and add one rule: the robot may not enter an obstacle row until aligned with the gap. A logged 150 s run is collision-free at 0.64 m/s, on average 0.66 m from the ball, and a longer run reaches stage 17 collision-free. The ball passes straight through the obstacles, so the robot alone avoids them.

1. INTRODUCTION

Most robots that move through cluttered spaces rely on rich sensing (lidar, depth cameras, or motion capture) and a mapping-and-planning stack on top. We ask the opposite question: how far can a four-legged robot get through a dense obstacle corridor with only a single RGB camera on its head and a walking controller it did not train?

The robot chases a moving red ball, a stand-in for a goal to walk toward. The corridor is five metres wide and packed with rows of obstacles whose gaps tighten with distance. The walking comes from a frozen, pre-trained reinforcement-learning (RL) policy from the RSL-RL framework [1], used as a black box whose weights we never change; the ball tracking is plain colour vision. The planner, the safety rules, the corridor generator, and the dashboard are ours.

Contribution. The policy of Rudin et al. [1] is omnidirectional: its command is a forward speed, a sideways speed, and a turn rate (v_x , v_y , v_ω). A controller that only walks toward the ball sets the forward speed and turn rate and leaves v_y at zero, so the robot is non-holonomic and must turn, walk, and turn back to shift sideways. We show that driving this unused sideways channel, together with a gate that refuses to enter a gap before the body is aligned, lets an untrained walker thread a tight corridor with no collisions at all. We did not find this combination, reusing the spare command channel of a frozen policy purely for reactive single-camera avoidance, in the literature we surveyed. Section 5 quantifies how much each layer contributes.

2. RELATED WORK

Learned legged locomotion. RL has produced strong quadruped controllers in simulation and on real hardware [2], including blind

walking over rough terrain [3] and policies that learn velocity commands in minutes [1]. We treat one such policy as a black box mapping a velocity command to joint motion, and never retrain it.

Reactive navigation. We build no map. The planner reads the next row of obstacles, picks a gap, and steers to it with a pure-pursuit law [4] — closer to follow-the-gap and potential-field methods than to global planning, which suits a low-sensing robot that must react fast.

Tracking the ball. The ball is found with an HSV colour mask and followed with CamShift [5], a light histogram tracker chosen over a neural detector to keep the pipeline simple to inspect. The simulator is MuJoCo [6] and the Go2 model is from the MuJoCo Menagerie.

3. SYSTEM DESIGN

The system wraps three layers around the frozen walking policy (Fig. 1). The camera feeds a tracker, the tracker feeds a planner, the planner and a hard gate shape a velocity command, and the frozen policy turns that command into a gait; only the robot's base pose and proprioception flow back.

- **Perception** (`tracker.py`) finds the red ball with an HSV mask and follows it with CamShift, re-detecting when confidence drops.
- **Reactive planning** (`controller.py`, `main.py`) picks a gap through the next row, commits to it, and converts it into a body-frame velocity command including the sideways slide. Pure pursuit follows the path; a hard gate and a push-away term are the safety net.
- **Walking** (`low_level.py`) is the frozen policy. It reads a 45-number observation plus the velocity command and outputs 12 joint targets at 50 Hz, tracked by a PD loop at 500 Hz.

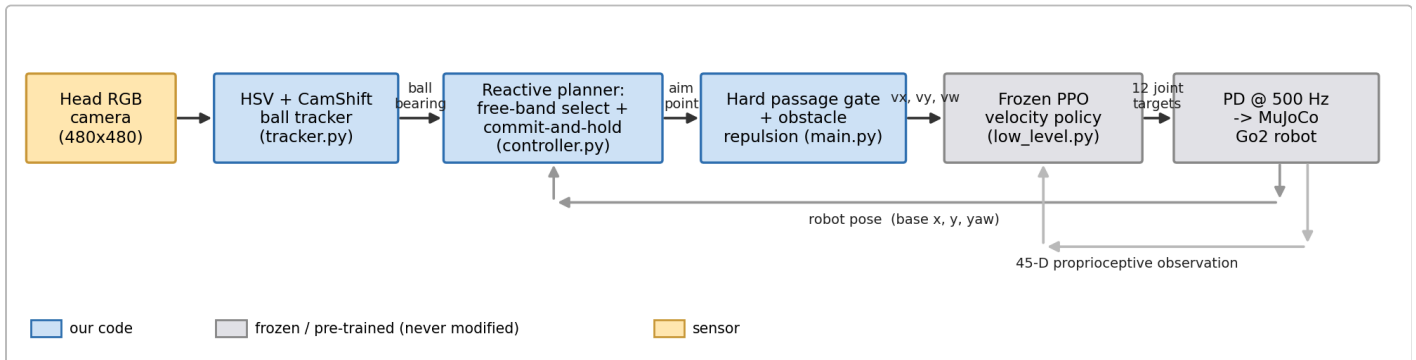


Fig. 1. System pipeline. Blue blocks are code we wrote; grey blocks are the frozen, pre-trained policy and the physics we never modify. The camera produces a ball bearing, the planner produces an aim point, the gate shapes the velocity command (v_x , v_y , v_ω), and only the robot's pose and a 45-D proprioceptive observation feed back.

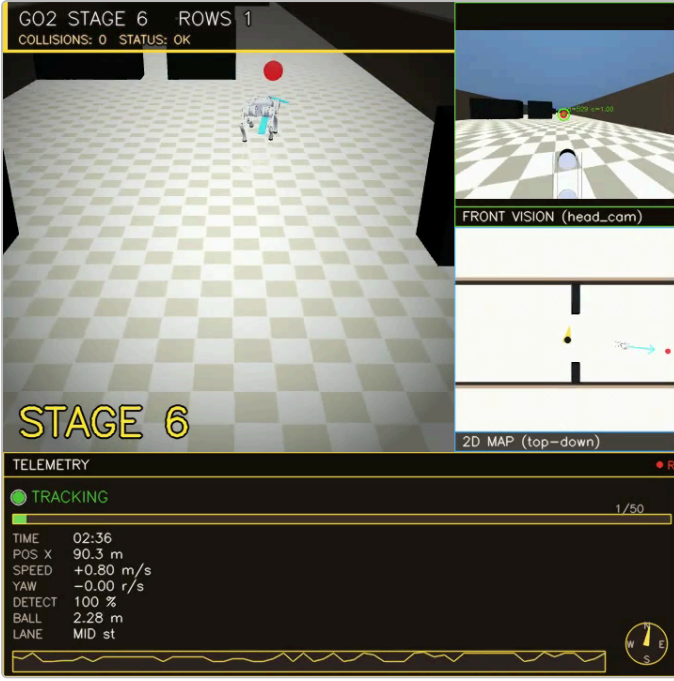


Fig. 2. The 720×720 dashboard. Left: the chase view with the planned path drawn on the floor and a halo on the obstacle being avoided. Top right: the robot's own head camera with the tracked ball. Below it: a top-down map. The bottom strip shows speed, range, detection confidence, a status light, and the stage.

A separate corridor generator lays out the obstacle rows, ramps difficulty with distance, and recycles passed rows to the front so the course never ends.

4. IMPLEMENTATION

4.1 Finding the ball

Red wraps around the hue circle, so we use two HSV ranges and clean the mask with morphology. The largest sensible blob seeds a box and a hue histogram, which CamShift tracks frame to frame. Short occlusions are tolerated because the tracker coasts; a long loss triggers a fresh detection. In simulation the ball's true position is known, so the controller keeps chasing while the ball is hidden behind an obstacle.

4.2 The planner and the gate

For the next row, the planner inflates each obstacle by a safety margin (about the robot's half-width plus slack), discards the blocked strips, and keeps the widest gap it can reach. It aims at the centre of that gap, never an edge, and stays committed until the robot is past the row; without commitment the ball can pull the robot back into an obstacle it just cleared.

The gate (Fig. 3) is the key mechanism. Until the robot is inside the chosen band it may not pass the front of the row: its forward speed drops to zero at that line while a strong sideways command slides it to the band centre, and only once aligned does it move through. Forcing alignment before entry keeps the robot safe wherever the ball leads it.

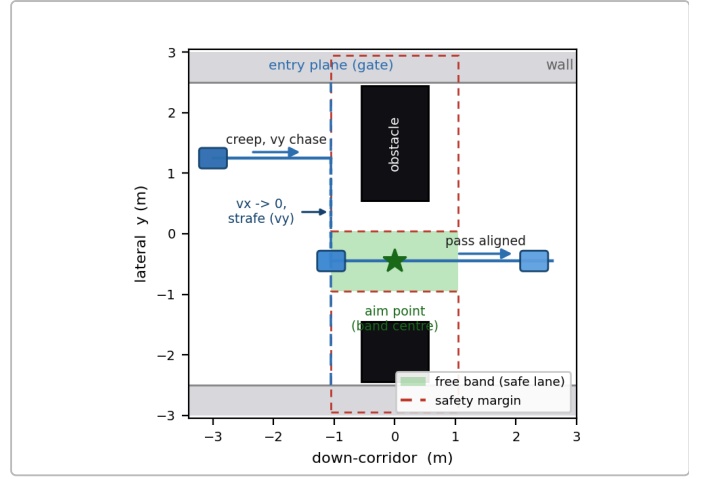


Fig. 3. The free-band passage gate (top-down). Each obstacle is inflated by a safety margin (dashed); the gap left between the inflated obstacles is the free band (green). The robot creeps to the entry plane, drops v_x to zero and strafes (v_y) to the band centre, then passes through aligned. It never crosses the gate while misaligned.

4.3 Turning the sideways channel back on

We rotate the high-level command into the robot's own frame and feed its sideways part into the policy's v_y input, which our first version left at zero. With it on, the robot steps directly into a wall-side gap instead of turning, walking, and turning back. A push-away term nudges the body sideways near an obstacle edge as a backup. We cap forward speed during sharp turns, where the frozen policy becomes unreliable, and a tripped robot stands up where it fell rather than restarting the run.

4.4 The corridor and the difficulty curve

Every row leaves at least one lane wide enough for the robot; otherwise two wide obstacles could split the corridor into impassable slivers and the planner would stall. Difficulty grows with distance, not time (Fig. 4): obstacles widen, the guaranteed lane narrows toward a floor of 1.30 m, and the colour darkens, across 17 stages that climb and then hold. The 1.30 m floor stays above the robot's 0.68 m width, so every row is passable but the later ones leave little room. Passed rows recycle to the front, making the corridor effectively endless, and the ball follows its own sine path straight through the obstacles, so the robot can never trail a cleared route.

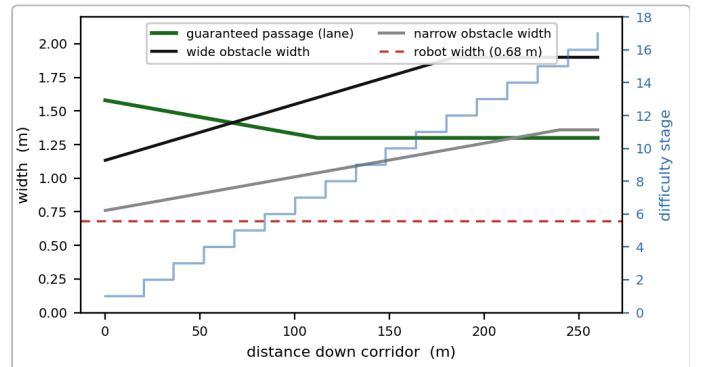


Fig. 4. Difficulty as a function of distance, straight from the generator. Obstacles widen while the guaranteed passage narrows to a 1.30 m floor, always above the robot's 0.68 m width (red line), so rows stay passable but tight. The blue staircase is the difficulty stage (1 to 17).

4.5 Control loop and runtime

The control loop runs at 50 Hz and the physics at 500 Hz: the policy reads one observation and emits one command every 20 ms while the PD loop and the contacts update ten times in between. With the scene trimmed to the obstacles in use, the simulator steps at about

5,700 Hz on a laptop, several times faster than real time. A single OpenCV window renders the full dashboard of Fig. 2.

5. RESULTS

We evaluated the system headless, counting any contact between the robot and an obstacle or wall as a failure. A logged 150 s run (seed 42) crossed fifty obstacle rows collision-free and without a fall, and a separate longer run held to stage 17 — about 320 m, roughly ten minutes — still collision-free (Fig. 6). Table 1 summarises the logged run; Fig. 5 details its trajectory.

Table 1. Headless evaluation summary.

Metric	Result
Logged run (seed 42)	150 s, 50 rows, 0 collisions, 0 falls

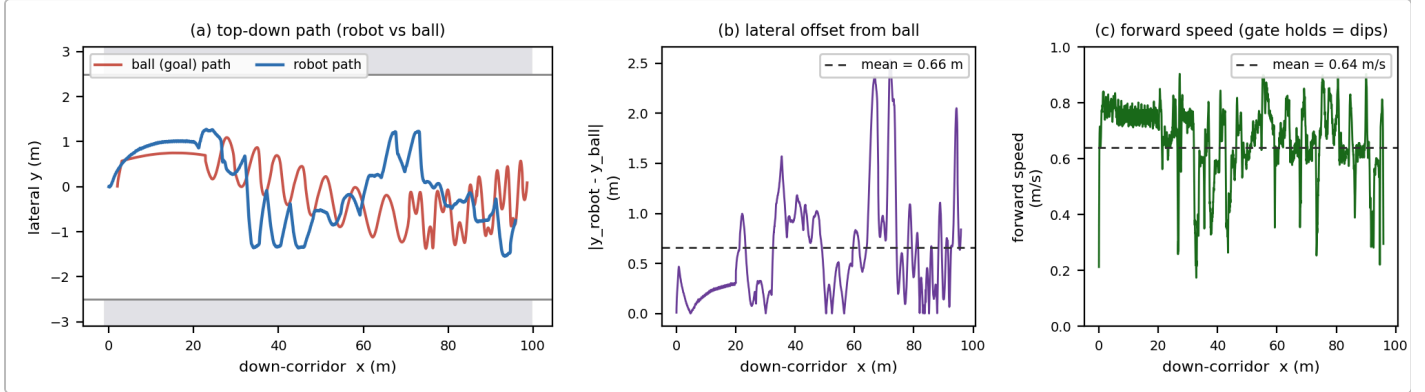


Fig. 5. One logged 150 s run (seed 42, collision-free). (a) Top-down paths: the robot (blue) weaves its own route while the ball (red) walks its sine path through the obstacles; grey bands are the corridor walls. (b) Lateral offset between robot and ball, mean 0.66 m. (c) Forward speed; the dips are gate alignment holds.

Robustness. To gauge how much the result leans on any single layer, we re-ran three seeds at maximum difficulty (every lane at the 1.30 m floor) with parts of the stack disabled. The robot stayed collision-free with the lateral-and-gate channel removed, and again with the reactive avoidance removed (Table 1). We read this as defence in depth: the guaranteed passable lane and the gap-steering planner already provide a safe baseline, so no single layer is a single point of failure. What the lateral-and-gate channel adds is precision rather than basic avoidance: with it on, the robot slows and centres itself in each gap (the speed dips in Fig. 5c) instead of cutting the corner; with it off, the robot clears the same rows but tracks straighter and faster.

Longest run (Fig. 6)	stage 17, ≈ 320 m, ≈ 10 min, 0 coll.
Mean forward speed	0.64 m/s (logged run)
Mean lateral offset	0.66 m (max 2.53 m $\approx$ width)
Ablation (3 seeds, max diff.)	collision-free, layers removed
Physics step rate	about 5,700 Hz

Fig. 5a shows the robot taking a different route from the ball: the ball weaves its own sine path and ploughs through obstacles while the robot detours around them. Their lateral offset (Fig. 5b) averages 0.66 m and spikes to the corridor half-width, confirming the robot avoids on its own rather than riding the ball's wake. Forward speed (Fig. 5c) averages 0.64 m/s and dips to near zero exactly when the gate is holding the robot to align it before a tight row.

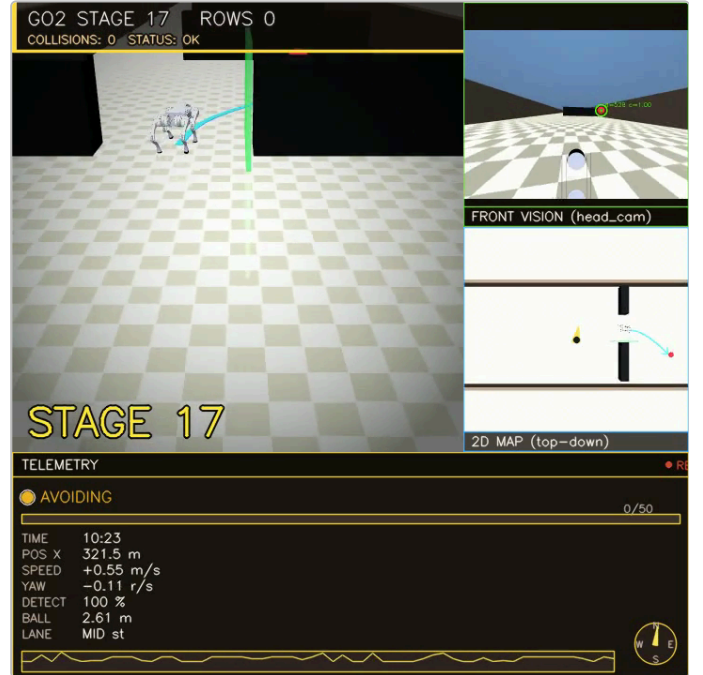


Fig. 6. Late in a run (stage 17, about 320 m in). Wide black obstacles fill the middle of the corridor and the robot threads the lane that is left; the green halo marks the obstacle being avoided, and the collision count is still zero.

6. DISCUSSION AND LIMITATIONS

Two failure modes shaped the design. First, the original controller drove only forward speed and yaw, so on a row whose only gap sat against a wall the robot could not line up and clipped the obstacle. This pushed us to drive the sideways channel and add the gate. Second, once the sideways channel was on, commanding a hard strafe and a hard turn together built up enough roll to tip the robot over. We fixed this by shrinking the sideways command during

sharp turns and capping forward speed in that part of the policy's command range. Both fixes are the lateral clip and the turn-rate guard of Section 4.3.

The walking policy is frozen and not fine-tuned for this task: it tracks sideways moves with about 0.25 m of error and trips occasionally during the hardest cross-corridor moves. We reduced both but did not remove them; the robot always recovered, and no trip became a collision in the runs we report. Vision is colour-based, so a change in lighting or ball colour would break it where a trained detector would not. Our evaluation covers a logged run plus a three-seed ablation rather than an exhaustive sweep. Next steps are to fine-tune the policy for tighter sideways tracking, replace the colour

tracker with a learned detector, and look one or two rows ahead so each gap is approached earlier.

7. CONCLUSION

A four-legged robot can cross a dense obstacle corridor collision-free — to stage 17, about 320 m, in a logged run — using only a head camera and a walking policy it never trained. Two changes made it work: drive the sideways velocity command the policy already exposes, and refuse to enter a gap before lining up with it. A robustness check shows the system leans on no single layer and degrades gracefully. The broader point is that extracting more from a general pre-trained policy can stand in for a good deal of custom control and extra sensing.

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